BAYVIEW AI CLUB

Can you fool AI?

Rock. Paper. Scissors.



Your move. Our model.

Where does our classifier fit?

AI

Systems that perform intelligent tasks

Machine learning

Learns patterns from examples

Deep learning

Neural networks with many layers

Our classifier

Two uses of AI

Classification
Chooses a label
“This is scissors.”

Generative AI
Produces new content
“Create a hand image.”

From pixels to a prediction

An image is a grid
of red, green and blue values.

Layers combine patterns
into useful image features.

Highest class score wins.

EXAMPLE ONLY

Rock 80% | Paper 15% | Scissors 5%



Training and inference

TRAINING

Labelled image
Pixels + answer

›

Prediction
Three class scores

›

Compare
How wrong was it?

›

Adjust
Update weights

Repeat with more examples

INFERENCE

New image › **Trained model** › **Predicted label**

P.S. The model's weights stay fixed during today's demo.

How do we know it works?

Training

Learn from examples

Validation

Choose during development

Testing

Check on separate examples

100%

Validation accuracy

89%

Test accuracy

Your hands? Let’s find out.

A confident prediction can still be wrong!

Why a good model can still fail

Overfitting
Training-specific clues.

New conditions
Lighting, backgrounds or hands.

High confidence
can still be wrong.

QUICK VOTE



Same gesture. Different conditions.

Would more of the same photos fix this?

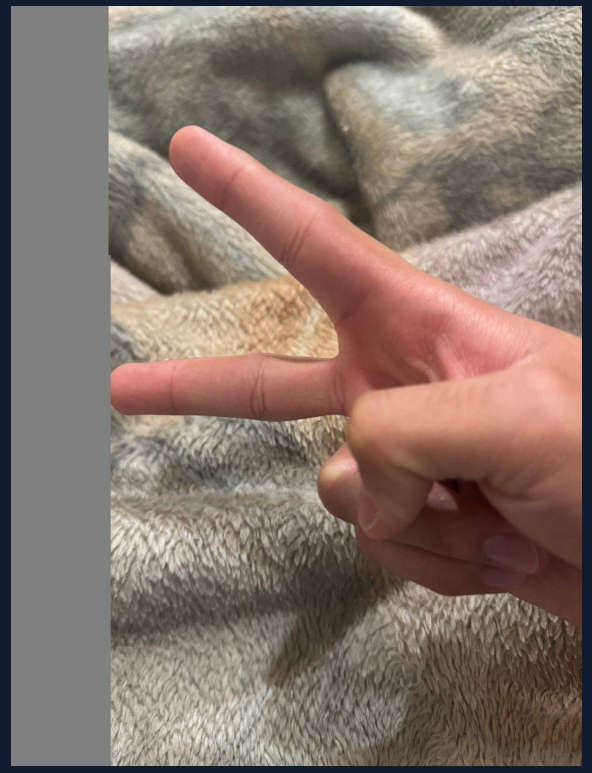
Will the model get it right?



A



B



C

Vote first. Then reveal the model’s prediction.

Challenge the classifier

Make a clear gesture the model misreads.

Angle **Background** **Lighting** **Distance**

1. Keep the gesture recognizable.
2. Change one thing at a time.
3. Predict the result before testing.

What explains the mistakes?

What changed?	Correct?	Possible explanation

What training examples would help?

One mistake gives us a clue. We still need to test it.

What did we learn?

Models learn patterns from examples.

Training changes the model.
Inference uses it.

New conditions can expose weaknesses.

NEXT MEETING

Your First ML Model